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Comparative study of rain measurement by optical disdrometers and tipping-bucket rain gauges in Basque Country

Santiago Gaztelumendi[1]; José Antonio Aranda[1]

[1] Basque Meteorology Agency (Euskalmet)

Corresponding author: Santiago Gaztelumendi, s-gaztelumendi@euskalmet.eus

Abstract:

This study examines and compares rainfall measurement characteristics from optical disdrometers (DIS) and tipping-bucket rain gauges (TBGs) across six different sites in the Basque Country over a period of three years. The main objective is to understand and quantify the conditions under which measurements from DIS and TBG are more or less comparable and representative. Various performance metrics are used to assess the agreement between DIS and TBG data, determining the dissimilarity and concordance of both methodologies in measuring precipitation. Limitations of using TBG records as a ground truth are acknowledged, with consideration given to hydrometeor classification and precipitation intensity. This study provides insight into the operational use of both instruments, their limitations, and potential biases.

1. Introduction

Precipitation measurement is crucial for managing extreme weather events, as it serves as a key indicator for natural disasters like droughts and flash floods. Ensuring accurate rainfall data is a priority worldwide. In the Basque Country, the autonomous government and local institutions are actively involved in maintaining a comprehensive observation network that includes rainfall recording.

Since its inception in 1990, the Basque Service of Meteorology (BSM) has been a reference point in the Basque Country for meteorological data services (e.g., GV 1990, 1997, 2022). Currently, the Basque Meteorology Agency (Euskalmet) remains at the forefront of operational meteorological affairs for the region, including the maintenance and enhancement of the Basque Automatic Weather Stations (AWS) network (e.g., Euskalmet 2024, Gaztelumendi et al. 2018).

The AWS network born in the late eighties with 20 AWS, has evolved during those last 30 years in different ways including better instrumentation, increasing progressively the density and covering the entire territory (Gaztelumendi et al 2018). In this moment, the Basque Country Mesonet measures more than 160 000 observations daily from its more than 120 AWS of different types each ten minutes (Euskalmet 2022).

The AWS network, which started in the late 1980s with 20 stations, has evolved over the past 30 years, incorporating better instrumentation and increasing density to cover the entire territory (Gaztelumendi et al. 2018). Currently, the Basque Country Mesonet records more than 160,000 observations daily from over 120 AWS of different types, with data collected every 10 minutes (Euskalmet 2022). The network is designed to measure mesoscale environmental variables, particularly of a hydrometeorological nature, and is managed by the Basque Government Directorate of Emergencies and

Meteorology (DAEM), the Basque Water Agency (URA), the Provincial Councils, and other public bodies (Gaztelumendi et al. 2018).

Within this mesonet (see Figure 1), rainfall has been observed every 10 minutes using TBGs for over 30 years (Gaztelumendi et al. 2003, 2018; Hernandez et al. 2022). Recently, optical Parsivel OTT disdrometers (OTT 2022, 2016) have been integrated into the network (Gaztelumendi et al. 2024), providing 1-minute rain rate measurements, among other variables (Gaztelumendi et al. 2022a, 2022b).

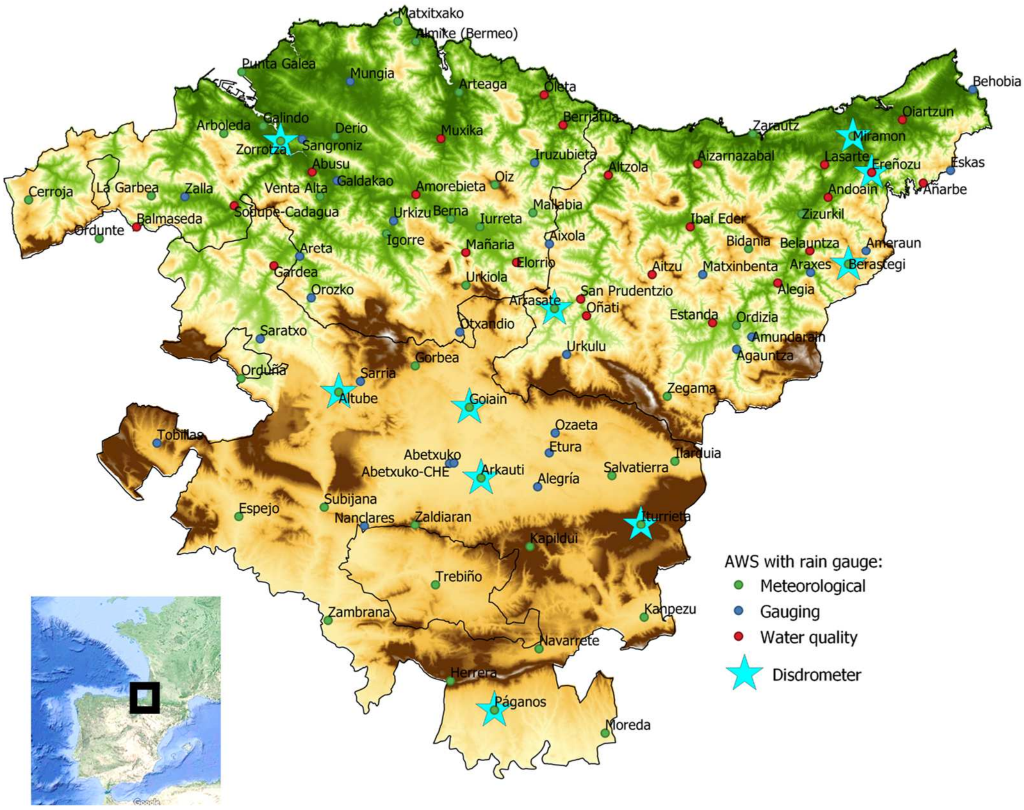
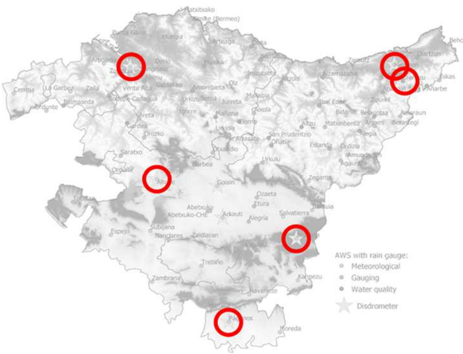


Figure 1: Disdrometers and tipping bucket rain-gauges in the Basque mesonetwork

2. Methods and data

The study covers observations from 2021 to 2023 obtained from DIS and TBGs at six sites (see Figure 2), where both DIS and TBG are present with a complete series of recorded data during this three-year period. To facilitate comparison, the DIS data series, which records precipitation in 1-minute intervals, was aggregated into 10-minute intervals to match the TBG series. In total, 289,002 records from TBGs and DIS were analyzed, representing 144,501 10-minute precipitation events across the six sites during the study period.



Name	alt (m)	lat	lon
Altube	618	42,9661	-2,86795
Ereñozu	25	43,242	-1,93922
Iturrieta	987	42,7935	-2,34575
Miramon	113	43,2868	-1,97121
Páganos	577	42,5605	-2,60055
Zorroza	5	43,2849	-2,96845

Figure 2: Disdrometers and tipping bucket rain-gauges selected for this study.

To aid in characterizing both datasets, precipitation is categorized by intensity, hydrometeor type, and other factors. Results are presented in tables and various graphical formats, facilitating the comparison of precipitation measurements under different segmentation scenarios.

Unlike traditional validation studies where one dataset serves as the reference or ground truth, in this analysis, neither TBG nor DIS is considered a definitive reference due to potential measurement uncertainties and inherent biases in both instruments. Instead, this study focuses on assessing the relative agreement, consistency, and differences between the two datasets using a variety of statistical and graphical methods. To ensure a fair comparison, DIS data were aggregated to match the temporal scales of TBG observations (10 minutes). Data quality control was performed by removing erroneous or missing values and applying basic quality checks.

Given the absence of a definitive reference dataset, a set of complementary metrics was selected to assess the comparability and agreement between TBG and DIS data. These metrics include measures of correlation, concordance, and relative error, which together provide a comprehensive assessment of both datasets. Descriptive statistics, histograms, and density plots provide a preliminary understanding of the data distributions and patterns. Subsequently, concordance and dissimilarity analyses are conducted to assess the strength and nature of the relationships between TBG and DIS data. Relative error metrics (rRMSE and rMAE) provide additional insights into the relative differences, while Bland-Altman plots help identify systematic biases and overall agreement levels. By employing a combination of these metrics, the study aims to achieve a robust understanding of the conditions under which TBG and DIS datasets are more or less comparable. For this purpose different R (R 2022) scripts have been developed and applied.

3. Results and discussion

A preliminary analysis of the concordance between DIS and TBG datasets is presented in Table 1, which shows a summary of some basic statistics comparing the precipitation events recorded by each instrument. For each dataset, we calculate the number of 10-minute precipitation events according to different criteria and express them as percentages relative to the total study period.

Table 1 indicates that more than 87% of the time, no precipitation is recorded by TBG, and more than 93% of the time, no precipitation is recorded by DIS. Thus, DIS registers 12% of precipitation events, while TBG registers only 6%, meaning DIS detects twice as many precipitation events as TBG. However, when considering only 10-minute events with precipitation greater than 0.1 mm, the percentage drops to 4.4% for DIS, below the 6% for TBG. This highlights that, in general, the DIS dataset contains more precipitation events if we consider all intensities, but this trend reverses when we exclude events with intensities less than 0.1 mm/10min. Additionally, the discrepancy where TBG does not record precipitation while DIS does is about 7%, but this proportion drops significantly to 0.41% when considering only events greater than 0.1 mm/10 minutes. This result is reasonable, considering the minimum resolution of TBG is 0.1 mm/10 min.

Table 1: Summary of some estatistics for general concordance betwen DIS and TBG.

% Prec	DIS=0	TBG=0	DIS!=0	DIS>=0.1	TBG!=0	DIS=0 TBG!=0	DIS<0.1 TBG!=0	TBG=0 DIS!=0	TBG=0 DIS>=0.1	TBG=0 DIS=0	TBG!=0 DIS!=0	TBG!=0 DIS>=0.1
Altube	83,48	93,63	16,52	5,51	6,37	0,41	1,58	10,57	0,72	83,07	5,96	1,58
Ereñozu	85,26	91,83	14,74	5,63	8,17	0,59	2,78	7,15	0,24	84,67	7,59	2,78
Iturrieta	88,18	94,41	11,82	3,94	5,59	1,38	2,37	7,61	0,71	86,80	4,21	2,37
Miramon	85,98	91,97	14,02	5,29	8,03	0,79	2,94	6,78	0,21	85,19	7,24	2,94
Páganos	92,58	97,46	7,42	1,94	2,54	0,48	0,90	5,36	0,30	92,10	2,07	0,90
Zorroza	88,90	94,46	11,10	3,76	5,54	0,92	2,09	6,48	0,31	87,98	4,63	2,09
tot	87,39	93,96	12,61	4,35	6,04	0,76	2,11	7,32	0,41	86,63	5,28	2,11

Furthermore, when analyzing the proportion of non-null precipitation events in DIS versus null events in TBG, we find that around 7% of cases record precipitation in DIS but not in TBG. If we focus on the subset of DIS events with more than 0.1 mm/10min, this proportion drops sharply to 0.41%. This confirms that, generally, the DIS dataset contains more precipitation events when considering all intensities recorded by the disdrometers; however, this would not hold true if we excluded events with less than 0.1 mm/10min (see Table 1)

The discrepancy between non-recorded precipitation in TBG and recorded precipitation in DIS is 7%, but it drops to just 0.5% when considering only events <=0.1 mm/10min. This is entirely reasonable given that the minimum resolution of the TBG is, obviously, 0.1 mm/10min (see Table 1).

These results are further supported by analyzing the distributions of the 10-minute data from both DIS and TBG. Figure 3 presents boxplots of the data distributions, showing that the overall data distributions are quite different. However, when focusing on 10-minute records with precipitation >=0.1 mm/10 min, the DIS and TBG series appear much more similar.

Figure 4 provides histograms for the DIS and TBG datasets segmented into different ranges of precipitation intensity. The left panel shows the distribution of events with intensities between 0.1 and 3 mm/10min, while the right panel shows the distribution for intensities between 0 and 0.1 mm/10min. The data reveal that the DIS dataset contains a wide variety of records between 0 and 0.1 mm/10min due to its original one-minute resolution, whereas the TBG dataset groups the values around 0 (no tipping) or 0.1 (one tip).

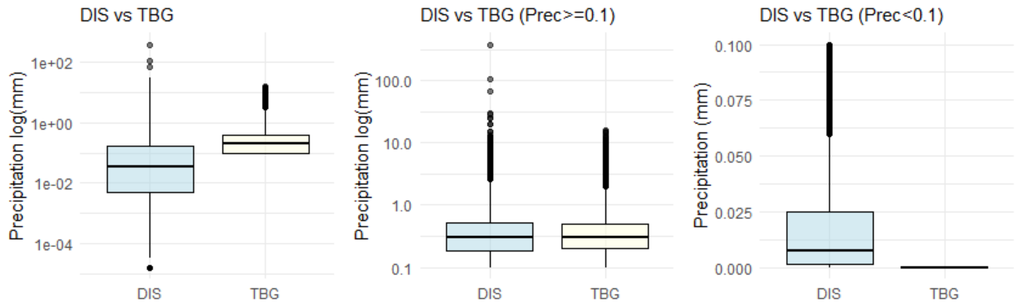


Figure 3: Boxplot of TBG and DIS data distributions for the full 10-minute dataset and segmented for >=0.1 y <0.1 mm/h.

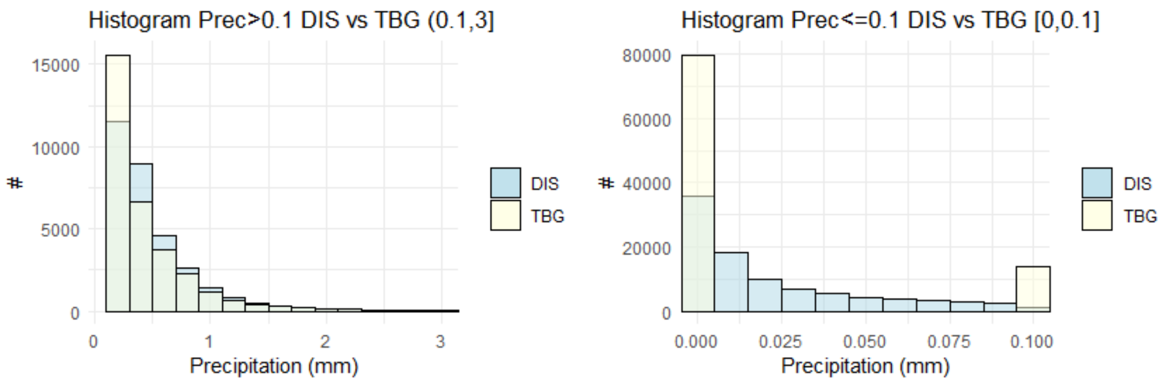


Figure 4: Histogram of the number of TBG and DIS events for different 10-minute precipitation ranges: (0.1-3] mm/10min (left) and 0,0.1) mm/10min (right).

Overall, the total precipitation recorded by DIS is about 15% higher than TBG. This discrepancy in precipitation totals, interpreted from the perspective of the TBG dataset, is distributed differently according to the range of intensities, types of hydrometeors, and other characteristics of each event. Figure 5 shows the distribution of events by hydrometeor type. The logarithmic scale and the total precipitation value indicated above each bar provide a clear understanding of the relative importance of each type of precipitation at each site, as well as its total contribution in terms of accumulated precipitation (mm).

To facilitate a comparative analysis at the event level, we generated a new variable representing the differences (diff) between TBG and DIS series for all available time points and locations. This difference series can be analyzed using various techniques. Figure 6 presents boxplots of the distributions of these differences segmented by different data subsets, allowing for a first qualitative analysis. The first graph in Figure 6 segments by hydrometeor type, showing that the differences are substantially larger when moving from liquid to solid phases, with smaller relative differences for cases categorized as "Rain." The second graph reveals that the differences increase with intensity (obviously), with TBG registering higher amounts than DIS in the range of 20 to 60 mm/h. The third graph confirms the expected behavior: the differences between DIS and TBG precipitation increase at lower temperatures, favoring DIS.

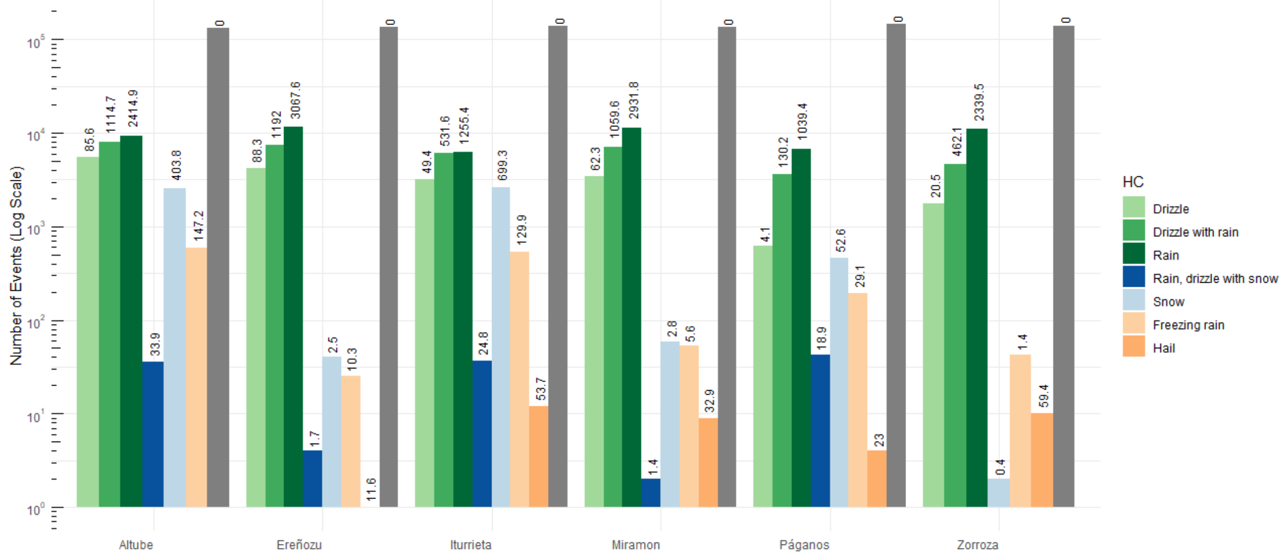


Figure 5: Number of events and total precipitation by location and precipitation type.

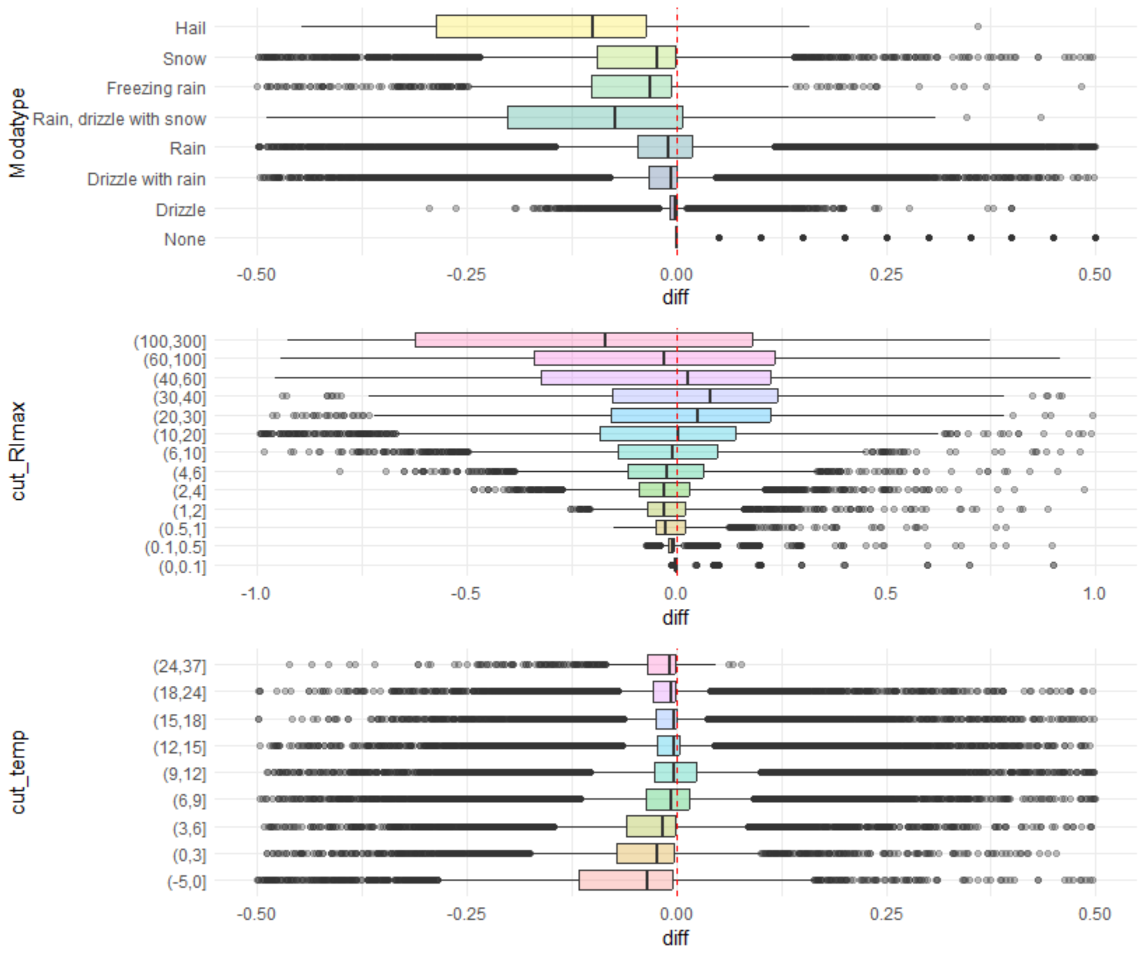


Figure 6: Boxplot for the difference TBG precipitation – DIS precipitation (diff) segmented by Hidrometeor class (Modatype), Maximum rain intensity classes (cut_Rimax), and temperature classes (cut_temp).

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Having reviewed the general behavior and some qualitative aspects of the differences, we now proceed to a quantitative analysis of the performance of both sensors under different real-world operating conditions. It is important to note that both zero values and rare extreme high values are present in the datasets. In this context, the usual correlation metrics (e.g., Pearson) applied to the complete dataset can provide conflicting information, as correlation does not capture differences in magnitude (bias) or variability, which are crucial for understanding how the two datasets compare. To avoid this problem, combined Metrics for Systematic Differences and Variability, particularly Lin's Concordance Correlation Coefficient (CCC) and the Adaptive Dissimilarity Index (ADI) was used. We also include additional relative error metrics like relative Root Mean Square Error (rRMSE) and relative Mean Absolute Error (rMAE), in Figure 7 we can the values of the commented metrics.

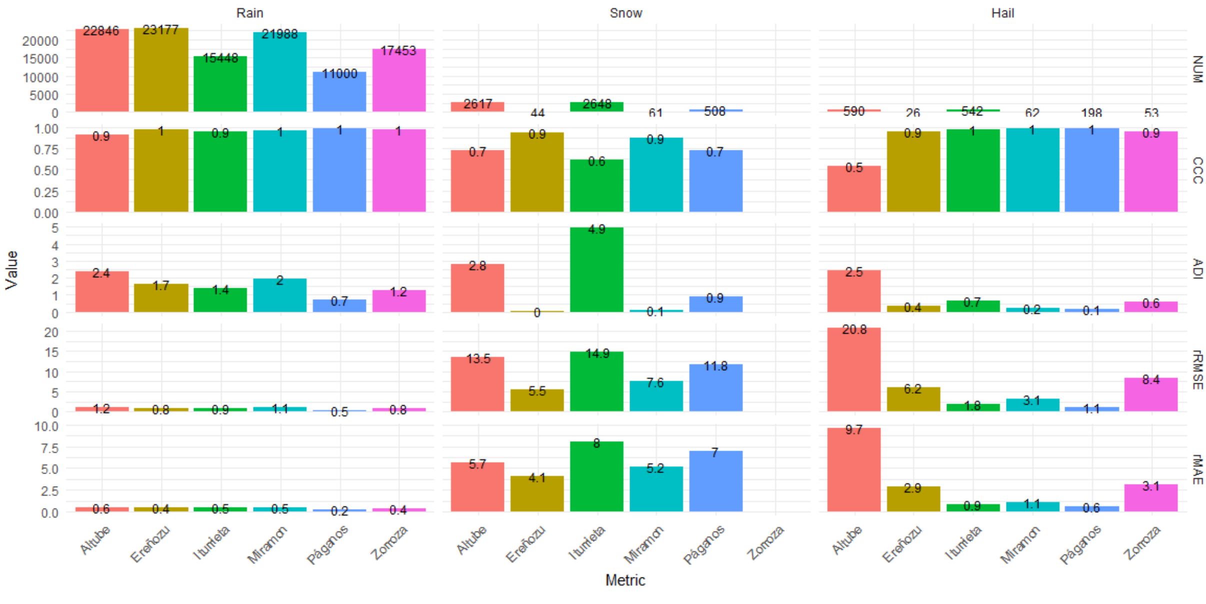

Figure 7: Bargraph with metric values for each location segmented by main HC classes (Rain, Snow, Hail) (NUM=number events considered)

The Concordance Correlation Coefficient (CCC) (Lin 1989) is a valuable statistic for assessing agreement and reliability between two sets of continuous measurements. Is a metric of both consistency/correlation/precision and accuracy. We also use an adaptive dissimilarity index ADI (Chouakria-Douzal and Nagabhushan 2007) that covers both dissimilarity on raw values and dissimilarity on temporal correlation behaviors using for the raw data discrepancy the "euclid" method. Thus, we can characterize the behavior of both series for each station by quantifying the concordance using the CCC and the dissimilarity using the ADI.

For field campaigns comparing different rain gauges (e.g., WMO 2009), relative difference vs. rain intensity graphs are often used, assuming a reference series considered the ground truth. In our case, this is not feasible, so we use Bland-Altman plots (Bland and Altman 1986) to analyze the difference between two measurements against their mean. This method is effective for evaluating the agreement between two measurement techniques. Consistent differences between TBG and DIS measurements are reflected as a shift from the zero line in the Bland-Altman plot. We can also identify heteroscedasticity (i.e., non-constant variance) between the measurements. For example, the spread of differences increases with the magnitude of rainfall, suggesting that the agreement between TBG and DIS varies with rainfall intensity. In the Bland-Altman plot, we highlight outliers where the two instruments disagree significantly. We combine the Bland-Altman plot by coloring the dots according to different characteristics to extract better conclusions. Figure 8 provides an example of how this method can be employed for more nuanced analysis.

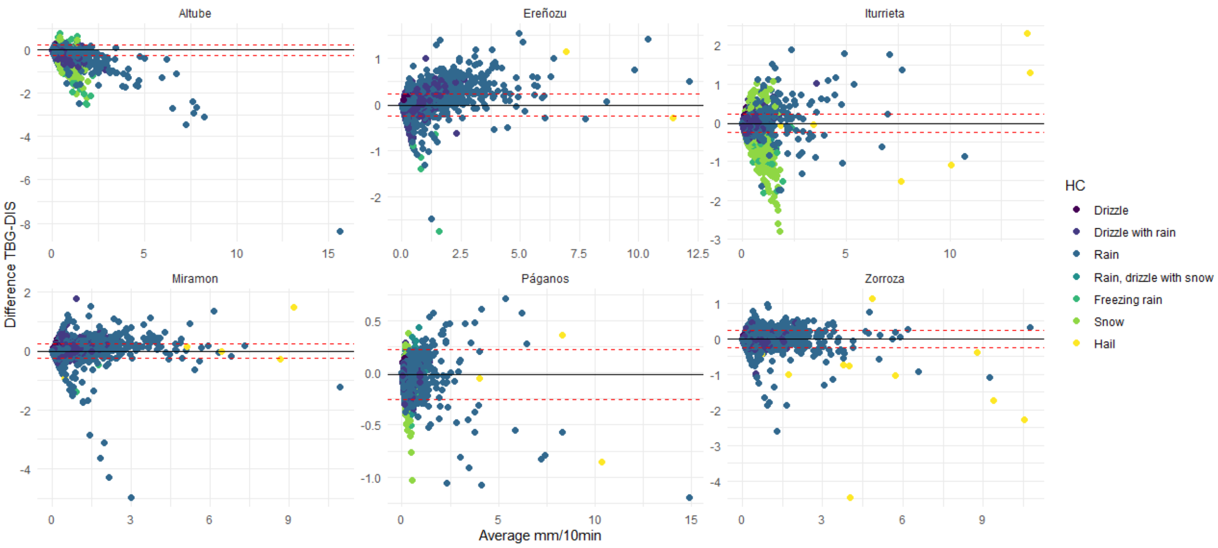

Figure 8: Bland-Almann plot for differences and mean DIS-TBG for ech location and Hydrometeor Class.

4. Conclusions

A preliminary methodology and set of tools have been developed to systematically compare rainfall data series from tipping bucket rain gauges (TBG) and disdrometers (DIS). These methods and tools will be applied to all available data series in future work and could be valuable for supporting other quality control processes (Hernandez et al. 2022).

By segmenting the total dataset based on location, hydrometeor types recorded by DIS, maximum precipitation intensity recorded by DIS, or temperature at the station, the study aims to provide a nuanced understanding of the comparability and reliability of these two instruments under different meteorological conditions.

Overall, the analyses suggest that while DIS and TBG generally exhibit substantial agreement under certain conditions (e.g., moderate rainfall intensities), significant differences arise under specific circumstances, such as low precipitation intensities or different solid hydrometeor types. This variability must be carefully considered in applications that rely on accurate precipitation measurements and extremes, particularly those requiring high temporal resolution or comprehensive hydrometeor classification.

Further work is required to include case studies, particularly during extreme events, and to conduct specific analyses of outliers. An event-based analysis seems necessary for a deeper understanding of sensor performance from a high temporal resolution perspective. Additionally, a more in-depth study is planned to better understand precipitation phenomena from an aggregated perspective of rainfall episodes (Gaztelumendi et al. 2024).

5. Acknowledgment

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